Term 3 — Lesson 1 Practice Activity (C5) with Solutions

Contents

Evaluated criteria

- C5: Calculates hypothesis tests on the mean of a population in counting contexts.

1 General Problem (One-Sample Mean Test Template in Counting Contexts)

A benchmark claim concerns a population mean count. A random sample of size n has x successes, so $\bar{x} = x/n$. At significance level α, test whether the true mean count is lower than, higher than, or different from the benchmark μ_0.

Reference table for common significance levels

Significance level	**Upper-tail p-value**	z_α **such that** $P(Z > z_\alpha) = \alpha$
10%	0.10	1.282
5%	0.05	1.645
2%	0.02	2.054
1%	0.01	2.326

Questions/Tasks.

(a) Identify the population parameter and write hypotheses for lower-tail, higher-tail, and two-tailed alternatives.

(b) State the condition checks for a one-sample mean count z-test.

(c) Write the one-sample mean count test statistic.

(d) State how the p-value is computed for lower-tail, higher-tail, and two-tailed tests.

(e) State the decision rule and write a contextual conclusion template.

C5

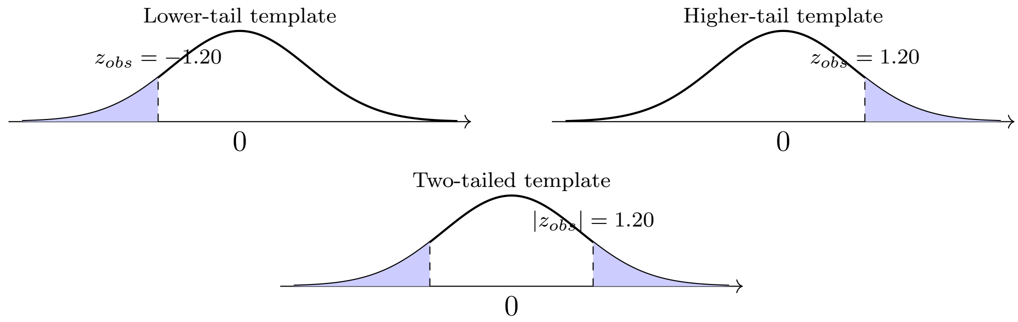
- **Parameter:** $\mu =$ true population mean count for the context.
- **Hypotheses:**

$$H_0\colon \mu = \mu_0, \qquad H_a\colon \mu < \mu_0 \text{ or } \mu > \mu_0 \text{ or } \mu \neq \mu_0.$$

- **Conditions:** Check $n\mu_0 \geq 10$ and $n(1 - \mu_0) \geq 10$.
- **Test statistic:**

$$z = \frac{\bar{x} - \mu_0}{\sqrt{\frac{\mu_0(1-\mu_0)}{n}}}.$$

- **p-value by tail type:**
  - Lower-tail: $P(Z \leq z_{obs})$
  - Higher-tail: $P(Z \geq z_{obs})$
  - Two-tailed: $2P(Z \geq |z_{obs}|)$
- **Decision rule:** Reject H_0 if p-value $\leq \alpha$; otherwise fail to reject H_0.
- **Conclusion template:** "There is sufficient evidence" (reject H_0) or "There is not enough evidence" (fail to reject H_0) for the contextual claim in H_a.



2 On-time assignment completion mean count (introductory higher-tail)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Consider two random samples:

- Sample A has 250 assignments, of which 190 are submitted on time.
- Sample B has 500 assignments, of which 380 are submitted on time.

Notice that both samples have the same sample mean count. At $\alpha = 5\%$, test whether the true on-time mean count is higher than 0.72 for each sample.

C5

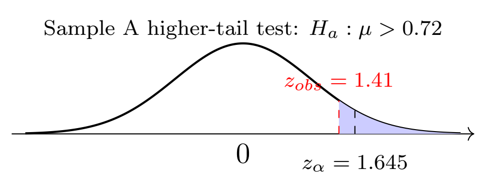
- **Parameter:** $\mu =$ true mean number of on-time submissions per assignment opportunity (0 or 1).
- **Hypotheses:** $H_0\colon \mu = 0.72$, $H_a\colon \mu > 0.72$.

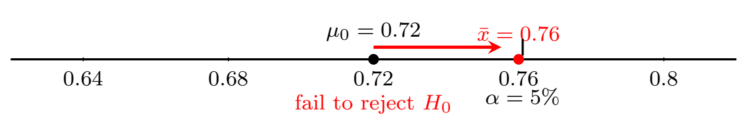
Sample A

- **Conditions:** $n\mu_0 = 250(0.72) = 180 \geq 10$, $n(1 - \mu_0) = 250(0.28) = 70 \geq 10$.
- **Sample mean count:** $\bar{x} = 190/250 = 0.76$.
- **Test statistic:**

$$z = \frac{0.76 - 0.72}{\sigma/\sqrt{250}} \approx 1.41, \quad \sigma = \sqrt{0.72(0.28)} \approx 0.4489.$$

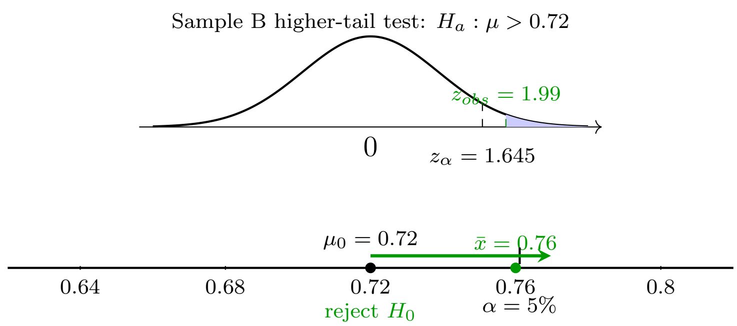
- **p-value:** $P(Z \geq 1.41) \approx 0.0795$.
- **Decision:** Fail to reject H_0 since $0.0795 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is above 0.72.





Sample B

- **Conditions:** $n\mu_0 = 500(0.72) = 360 \geq 10$, $n(1-\mu_0) = 500(0.28) = 140 \geq 10$.
- **Sample mean count:** $\bar{x} = 380/500 = 0.76$.
- **Test statistic:**
$$z = \frac{0.76 - 0.72}{\sigma/\sqrt{500}} \approx 1.99, \quad \sigma = \sqrt{0.72(0.28)} \approx 0.4489.$$
- **p-value:** $P(Z \geq 1.99) \approx 0.0232$.
- **Decision:** Reject H_0 since $0.0232 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission mean count is above 0.72.



Although both samples have the same sample mean count $\bar{x} = 0.76$, the larger sample gives a more positive test statistic and a smaller p-value because the standard error is smaller.

3 On-time assignment completion mean count (introductory lower-tail)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Consider two random samples:

- Sample A has 250 assignments, of which 170 are submitted on time.
- Sample B has 500 assignments, of which 340 are submitted on time.

Notice that both samples have the same sample mean count. At $\alpha = 5\%$, test whether the true on-time mean count is lower than 0.72 for each sample.

C5

- **Parameter:** $\mu =$ true mean number of on-time submissions per assignment opportunity (0 or 1).
- **Hypotheses:** $H_0\colon \mu = 0.72$, $H_a\colon \mu < 0.72$.

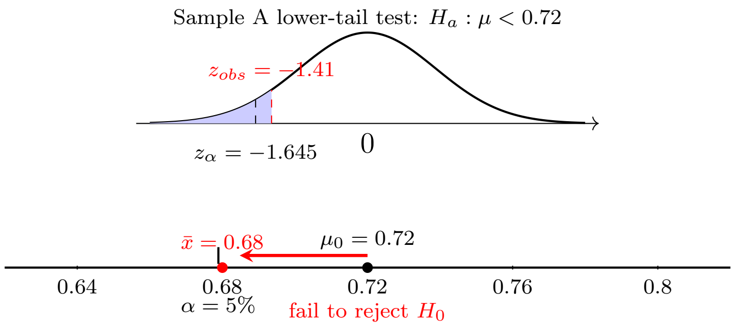
Sample A

- **Conditions:** $n\mu_0 = 250(0.72) = 180 \geq 10$, $n(1-\mu_0) = 250(0.28) = 70 \geq 10$.
- **Sample mean count:** $\bar{x} = 170/250 = 0.68$.

- **Test statistic:**

$$z = \frac{0.68 - 0.72}{\sqrt{\frac{\sigma^2}{250}}} \approx -1.41.$$

- **p-value:** $P(Z \leq -1.41) \approx 0.0795$.
- **Decision:** Fail to reject H_0 since $0.0795 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is below 0.72.

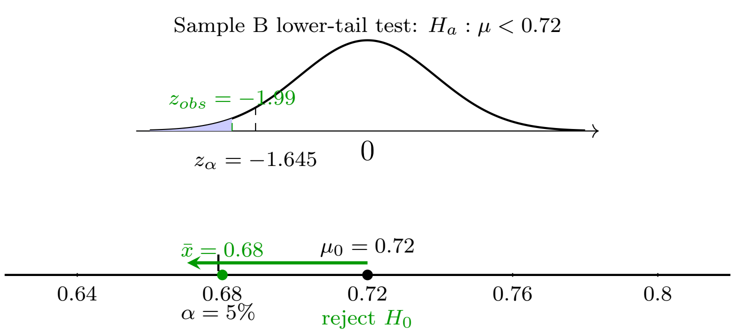


Sample B

- **Conditions:** $n\mu_0 = 500(0.72) = 360 \geq 10$, $n(1 - \mu_0) = 500(0.28) = 140 \geq 10$.
- **Sample mean count:** $\bar{x} = 340/500 = 0.68$.
- **Test statistic:**

$$z = \frac{0.68 - 0.72}{\sqrt{\frac{\sigma^2}{500}}} \approx -1.99.$$

- **p-value:** $P(Z \leq -1.99) \approx 0.0232$.
- **Decision:** Reject H_0 since $0.0232 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission mean count is below 0.72.



Although both samples have the same sample mean count $\bar{x} = 0.68$, the larger sample gives a more negative test statistic and a smaller p-value because the standard error is smaller.

4 On-time assignment completion mean count (introductory two-tailed)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Consider two random samples:

- Sample A has 250 assignments, of which 190 are submitted on time.
- Sample B has 500 assignments, of which 380 are submitted on time.

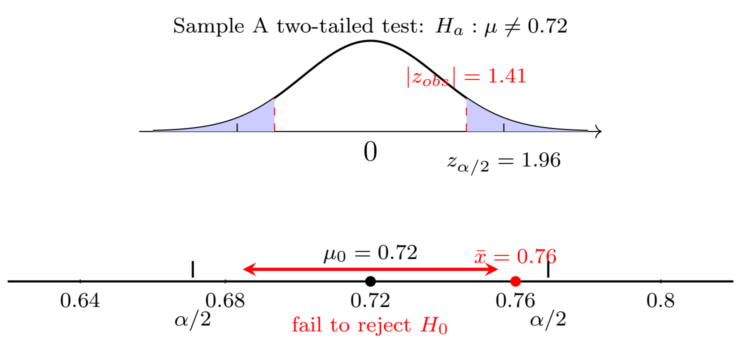
Notice that both samples have the same sample mean count. At $\alpha = 5\%$, test whether the true on-time mean count is different from 0.72 for each sample.

C5

- **Parameter:** $\mu =$ true mean number of on-time submissions per assignment opportunity (0 or 1).
- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.72$, $H_a\colon \mu \neq 0.72$.

Sample A

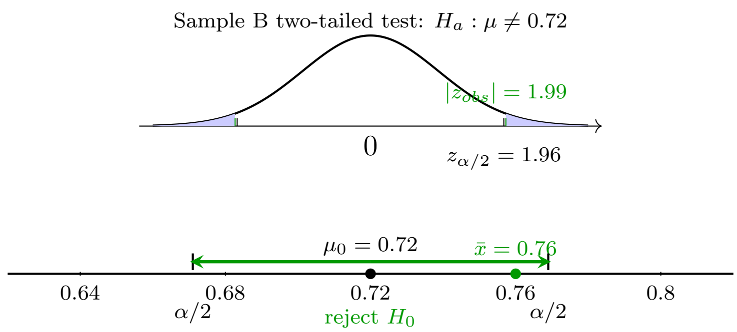
- **Conditions:** $n\mu_0 = 250(0.72) = 180 \geq 10$, $n(1-\mu_0) = 250(0.28) = 70 \geq 10$.
- **Sample mean count:** $\bar{x} = 190/250 = 0.76$.
- **Test statistic:**
$$z = \frac{0.76 - 0.72}{\sigma/\sqrt{250}} \approx 1.41, \quad \sigma = \sqrt{0.72(0.28)} \approx 0.4489.$$
- **p-value:** $2P(Z \geq 1.41) \approx 0.1590$.
- **Decision:** Fail to reject H_0 since $0.1590 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is different from 0.72.



Sample B

- **Conditions:** $n\mu_0 = 500(0.72) = 360 \geq 10$, $n(1-\mu_0) = 500(0.28) = 140 \geq 10$.
- **Sample mean count:** $\bar{x} = 380/500 = 0.76$.
- **Test statistic:**
$$z = \frac{0.76 - 0.72}{\sigma/\sqrt{500}} \approx 1.99, \quad \sigma = \sqrt{0.72(0.28)} \approx 0.4489.$$

- **p-value:** $2P(Z \geq 1.99) \approx 0.0464$.
- **Decision:** Reject H_0 since $0.0464 < 0.05$.
- **Conclusion:** There is enough evidence that the true on-time submission mean count is different from 0.72.



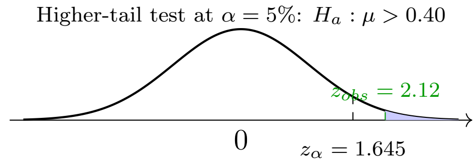
Although both samples have the same sample mean count $\bar{x} = 0.76$, the larger sample gives a larger absolute test statistic and a smaller p-value because the standard error is smaller.

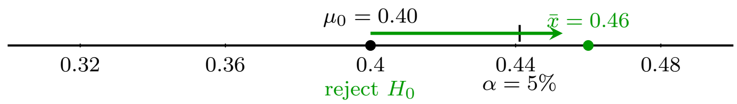
5 Platform adoption rate after training (higher-tail test compared across three significance levels)

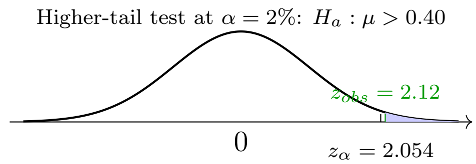
A district dashboard states that teachers average 0.40 regular uses of a new digital platform per observation period (counting variable: 1 if used regularly, 0 otherwise). After training, a random sample of 300 teachers shows 138 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true adoption mean count is higher than 0.40.

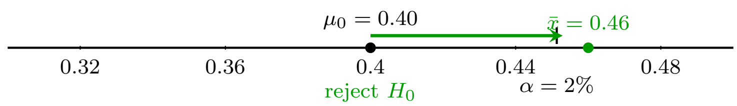
C5

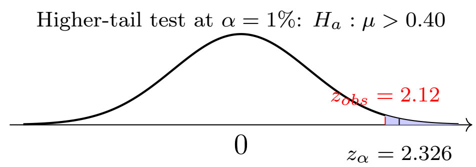
- **Parameter:** μ = true mean number of regular platform uses per observation period (0 or 1).
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0\colon \mu = 0.40$, $H_a\colon \mu > 0.40$.
- **Conditions:** $n\mu_0 = 120 \geq 10$, $n(1-\mu_0) = 180 \geq 10$.
- **Sample mean count:** $\bar{x} = 138/300 = 0.46$.
- **Test statistic:**
$$z = \frac{0.46 - 0.40}{\sqrt{\frac{\sigma^2}{300}}} \approx 2.12.$$
- **p-value:** $P(Z \geq 2.12) \approx 0.0170$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0170 < 0.05$.
- **Decision at $\alpha = 2\%$:** Reject H_0 since $0.0170 < 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0170 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% and 2% significance levels, but it is not rejected at the 1% significance level. Therefore, only one of the three significance levels fails to reject H_0.

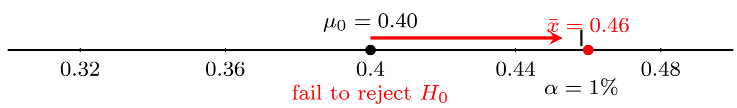












6 Platform adoption rate after training (lower-tail test compared across three significance levels)

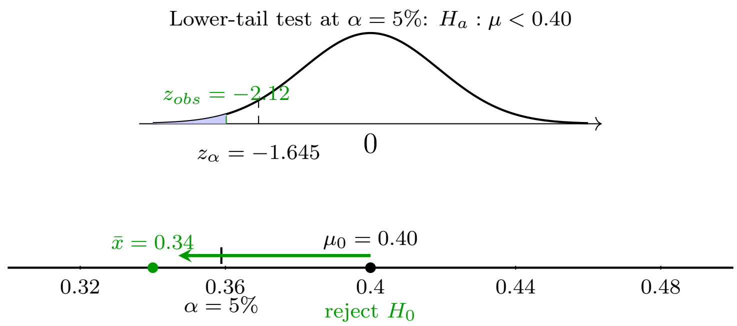
A district dashboard states that teachers average 0.40 regular uses of a new digital platform per observation period (counting variable: 1 if used regularly, 0 otherwise). After training, a random sample of 300 teachers shows 102 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true adoption mean count is lower than 0.40.

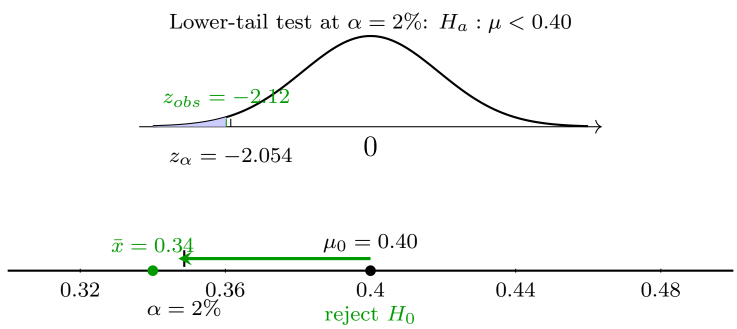
C5

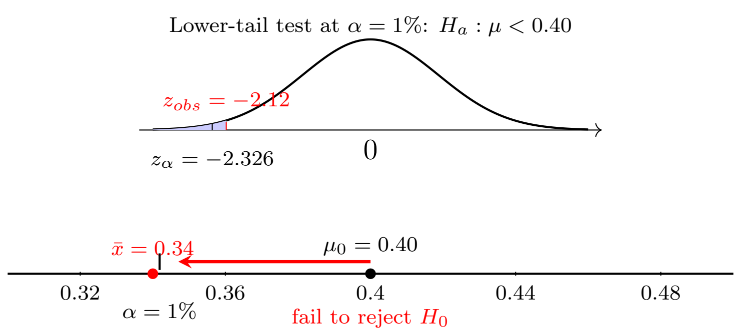
- **Parameter:** $\mu =$ true mean number of regular platform uses per observation period (0 or 1).
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0\colon \mu = 0.40$, $H_a\colon \mu < 0.40$.
- **Conditions:** $n\mu_0 = 120 \geq 10$, $n(1-\mu_0) = 180 \geq 10$.
- **Sample mean count:** $\bar{x} = 102/300 = 0.34$.
- **Test statistic:**

$$z = \frac{0.34 - 0.40}{\sqrt{\frac{\sigma^2}{300}}} \approx -2.12.$$

- **p-value:** $P(Z \leq -2.12) \approx 0.0170$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0170 < 0.05$.
- **Decision at $\alpha = 2\%$:** Reject H_0 since $0.0170 < 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0170 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% and 2% significance levels, but it is not rejected at the 1% significance level. Therefore, only one of the three significance levels fails to reject H_0.





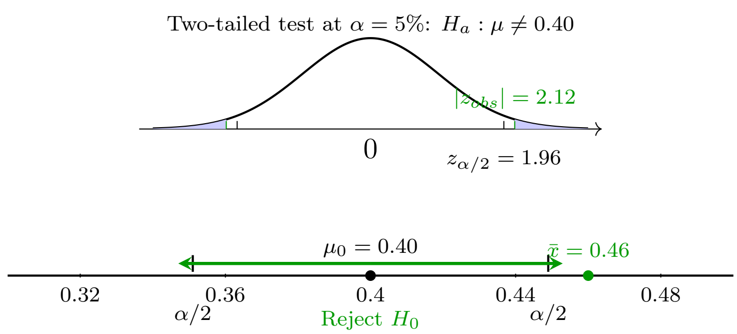


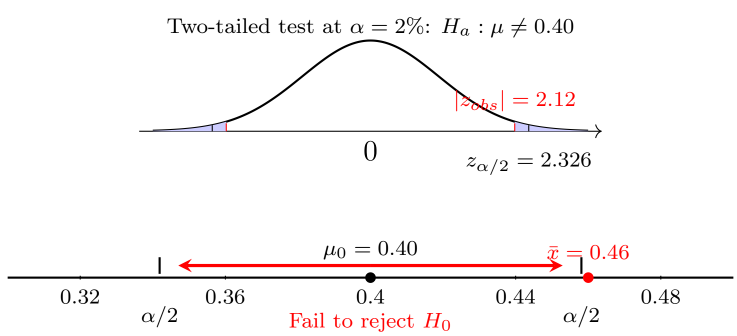
7 Platform adoption rate after training (two-tailed test compared across three significance levels)

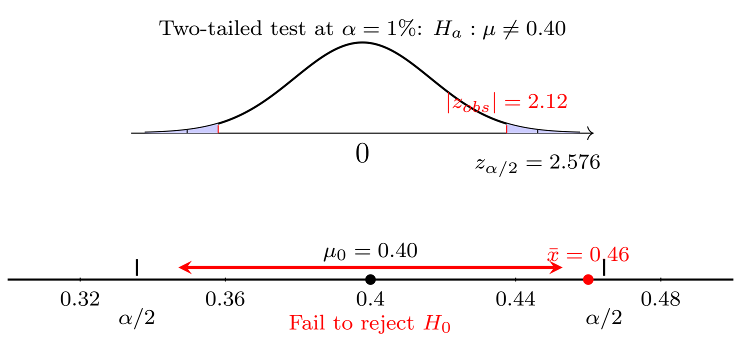
A district dashboard states that teachers average 0.40 regular uses of a new digital platform per observation period (counting variable: 1 if used regularly, 0 otherwise). After training, a random sample of 300 teachers shows 138 regular users. At significance levels $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$, test whether the true adoption mean count is different from 0.40.

C5

- **Parameter:** μ = true mean number of regular platform uses per observation period (0 or 1).
- **Hypotheses at $\alpha = 1\%$, $\alpha = 2\%$, and $\alpha = 5\%$:** $H_0\colon \mu = 0.40$, $H_a\colon \mu \neq 0.40$.
- **Conditions:** $n\mu_0 = 120 \geq 10$, $n(1-\mu_0) = 180 \geq 10$.
- **Sample mean count:** $\bar{x} = 138/300 = 0.46$.
- **Test statistic:**
$$z = \frac{0.46 - 0.40}{\sqrt{\frac{\sigma^2}{300}}} \approx 2.12.$$
- **p-value:** $2P(Z \geq 2.12) \approx 0.0340$.
- **Decision at $\alpha = 5\%$:** Reject H_0 since $0.0340 < 0.05$.
- **Decision at $\alpha = 2\%$:** Fail to reject H_0 since $0.0340 > 0.02$.
- **Decision at $\alpha = 1\%$:** Fail to reject H_0 since $0.0340 > 0.01$.
- **Conclusion:** The null hypothesis is rejected at the 5% significance level, but it is not rejected at the 2% and 1% significance levels.







8 On-time assignment completion mean count (comparison of lower-tail and higher-tail tests in two samples)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Consider two random samples:

- Sample A has 250 assignments, of which 168 are submitted on time.
- Sample B has 500 assignments, of which 377 are submitted on time.

For learning purposes, apply both one-sided tests to each sample, even though in Sample A the observed mean count is below 0.72 and in Sample B the observed mean count is above 0.72. At $\alpha = 5\%$, test whether the true on-time mean count is lower than 0.72 and whether it is higher than 0.72 for each sample.

C5

- **Parameter:** $\mu =$ true mean number of on-time submissions per assignment opportunity (0 or 1).

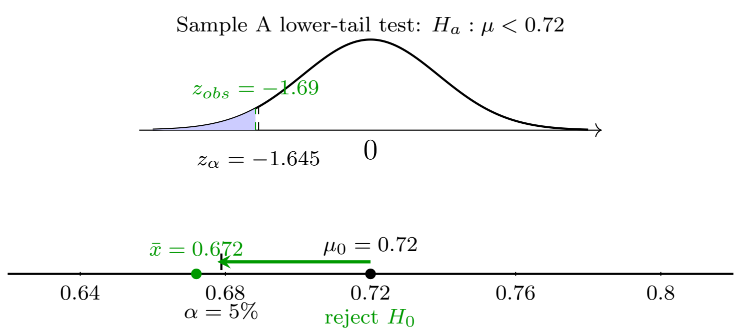
Sample A

- **Conditions:** $n\mu_0 = 250(0.72) = 180 \geq 10$, $n(1-\mu_0) = 250(0.28) = 70 \geq 10$.
- **Sample mean count:** $\bar{x} = 168/250 = 0.672$.
- **Test statistic:**

$$z = \frac{0.672 - 0.72}{\sqrt{\frac{\sigma^2}{250}}} \approx -1.69.$$

Test 1: Lower-tail test for Sample A

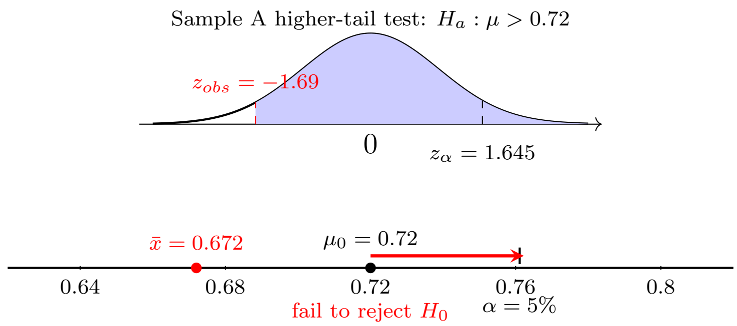
- **Hypotheses at $\alpha = 5\%$:** $H_0 \colon \mu = 0.72$, $H_a \colon \mu < 0.72$.
- **p-value:** $P(Z \leq -1.69) \approx 0.0455$.
- **Decision:** Reject H_0 since $0.0455 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true on-time submission mean count is below 0.72.



Test 2: Higher-tail test for Sample A

- **Hypotheses at $\alpha = 5\%$:** $H_0 \colon \mu = 0.72$, $H_a \colon \mu > 0.72$.

- **p-value:** $P(Z \geq -1.69) \approx 0.9545$.
- **Decision:** Fail to reject H_0 since $0.9545 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is above 0.72.



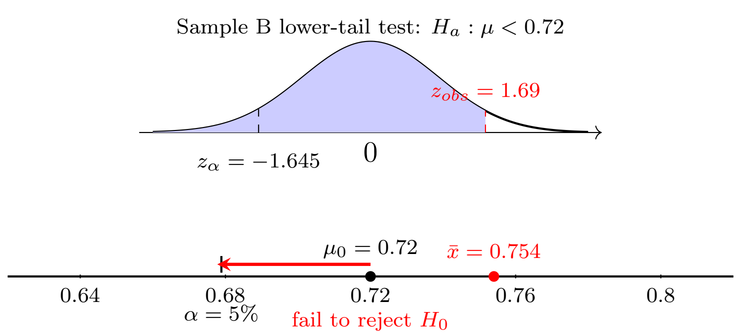
Sample B

- **Conditions:** $n\mu_0 = 500(0.72) = 360 \geq 10$, $n(1-\mu_0) = 500(0.28) = 140 \geq 10$.
- **Sample mean count:** $\bar{x} = 377/500 = 0.754$.
- **Test statistic:**

$$z = \frac{0.754 - 0.72}{\sqrt{\frac{\sigma^2}{500}}} \approx 1.69.$$

Test 1: Lower-tail test for Sample B

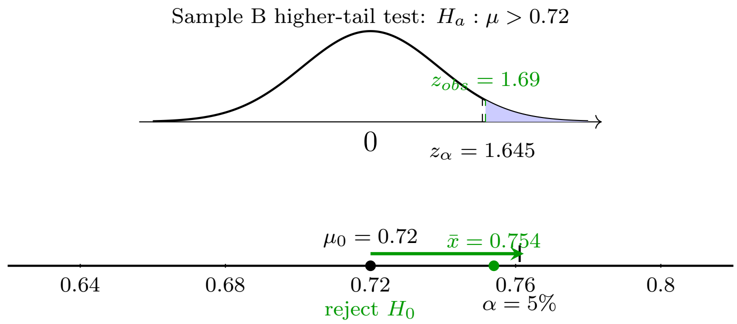
- **Hypotheses at $\alpha = 5\%$:** $H_0 \colon \mu = 0.72$, $H_a \colon \mu < 0.72$.
- **p-value:** $P(Z \leq 1.69) \approx 0.9545$.
- **Decision:** Fail to reject H_0 since $0.9545 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is below 0.72.



Test 2: Higher-tail test for Sample B

- **Hypotheses at $\alpha = 5\%$:** $H_0 \colon \mu = 0.72$, $H_a \colon \mu > 0.72$.
- **p-value:** $P(Z \geq 1.69) \approx 0.0455$.
- **Decision:** Reject H_0 since $0.0455 < 0.05$.

- **Conclusion:** There is sufficient evidence that the true on-time submission mean count is above 0.72.



In Sample A, the observed mean count is below the benchmark, so the lower-tail alternative is the direction supported by the data. In Sample B, the observed mean count is above the benchmark, so the higher-tail alternative is the direction supported by the data.

9 On-time assignment completion mean count (comparison of lower-tail and higher-tail tests in two samples, all fail to reject)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Consider two random samples:

- Sample A has 250 assignments, of which 173 are submitted on time.
- Sample B has 500 assignments, of which 372 are submitted on time.

For learning purposes, apply both one-sided tests to each sample, even though in Sample A the observed mean count is below 0.72 and in Sample B the observed mean count is above 0.72. At $\alpha = 5\%$, test whether the true on-time mean count is lower than 0.72 and whether it is higher than 0.72 for each sample.

C5

- **Parameter:** μ = true mean number of on-time submissions per assignment opportunity (0 or 1).

Sample A

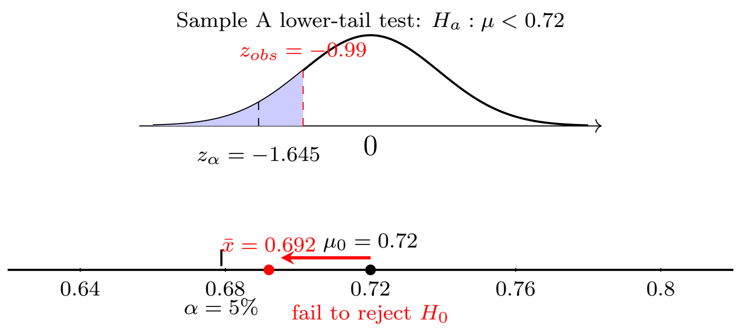
- **Conditions:** $n\mu_0 = 250(0.72) = 180 \geq 10$, $n(1-\mu_0) = 250(0.28) = 70 \geq 10$.
- **Sample mean count:** $\bar{x} = 173/250 = 0.692$.
- **Test statistic:**

$$z = \frac{0.692 - 0.72}{\sqrt{\frac{\sigma^2}{250}}} \approx -0.99.$$

Test 1: Lower-tail test for Sample A

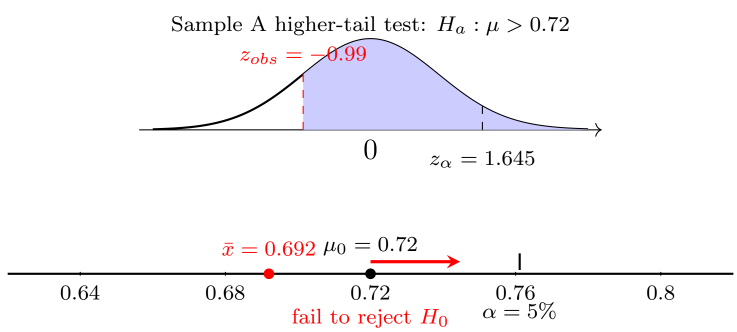
- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.72$, $H_a\colon \mu < 0.72$.
- **p-value:** $P(Z \leq -0.99) \approx 0.1611$.

- **Decision:** Fail to reject H_0 since $0.1611 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is below 0.72.



Test 2: Higher-tail test for Sample A

- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.72$, $H_a\colon \mu > 0.72$.
- **p-value:** $P(Z \geq -0.99) \approx 0.8389$.
- **Decision:** Fail to reject H_0 since $0.8389 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is above 0.72.



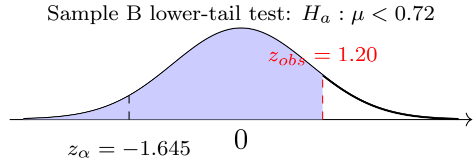
Sample B

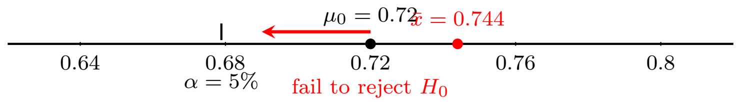
- **Conditions:** $n\mu_0 = 500(0.72) = 360 \geq 10$, $n(1-\mu_0) = 500(0.28) = 140 \geq 10$.
- **Sample mean count:** $\bar{x} = 372/500 = 0.744$.
- **Test statistic:**

$$z = \frac{0.744 - 0.72}{\sqrt{\frac{\sigma^2}{500}}} \approx 1.20.$$

Test 1: Lower-tail test for Sample B

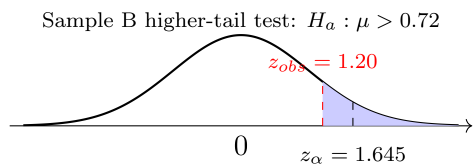
- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.72$, $H_a\colon \mu < 0.72$.
- **p-value:** $P(Z \leq 1.20) \approx 0.8849$.
- **Decision:** Fail to reject H_0 since $0.8849 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is below 0.72.

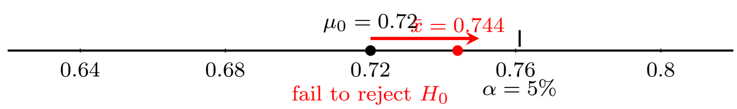




Test 2: Higher-tail test for Sample B

- **Hypotheses at $\alpha = 5\%$:** $H_0 \colon \mu = 0.72$, $H_a \colon \mu > 0.72$.
- **p-value:** $P(Z \geq 1.20) \approx 0.1151$.
- **Decision:** Fail to reject H_0 since $0.1151 > 0.05$.
- **Conclusion:** There is not enough evidence that the true on-time submission mean count is above 0.72.





In Sample A, the observed mean count is below the benchmark, so the lower-tail alternative is the direction supported by the data. In Sample B, the observed mean count is above the benchmark, so the higher-tail alternative is the direction supported by the data. However, in both samples the evidence is not strong enough at the 5% significance level, so all tests fail to reject H_0.

10 Compliance with safety checklist (Comparison of higher-tail and two-tailed I)

A quality team states that 0.85 of shifts complete a safety checklist. In a random sample of 320 shifts, 288 completed the checklist. At $\alpha = 5\%$, test both whether the true completion mean count is higher than 0.85 and whether the true completion mean count is different from 0.85.

C5

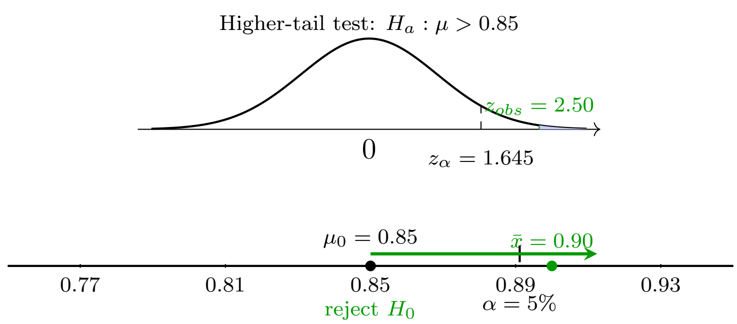
- **Parameter:** μ = true mean number of completed safety checklists per request (0 or 1).
- **Conditions:** $n\mu_0 = 272 \geq 10$, $n(1-\mu_0) = 48 \geq 10$.
- **Sample mean count:** $\bar{x} = 288/320 = 0.90$.

Test 1: Higher-tail test

- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.85$, $H_a\colon \mu > 0.85$.
- **Test statistic:**

$$z = \frac{0.90 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 2.50.$$

- **p-value:** $P(Z \geq 2.50) \approx 0.0062$.
- **Decision:** Reject H_0 since $0.0062 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion mean count is higher than 0.85.

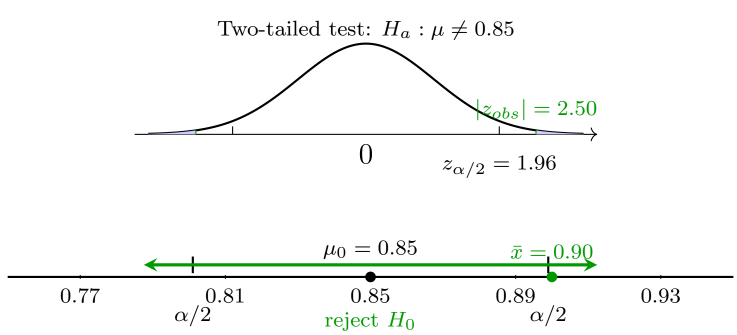


Test 2: Two-tailed test

- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.85$, $H_a\colon \mu \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.90 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 2.50.$$

- **p-value:** $2P(Z \geq 2.50) \approx 0.0124$.
- **Decision:** Reject H_0 since $0.0124 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion mean count is different from 0.85.



11 Compliance with safety checklist (Comparison of higher-tail and two-tailed II)

A quality team states that 0.85 of shifts complete a safety checklist. In a random sample of 320 shifts, 282 completed the checklist. At $\alpha = 5\%$, test both whether the true completion mean count is higher than 0.85 and whether the true completion mean count is different from 0.85.

C5

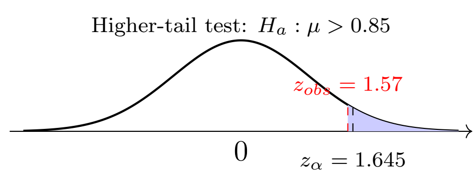
- **Parameter:** μ = true mean number of completed safety checklists per request (0 or 1).
- **Conditions:** $n\mu_0 = 272 \geq 10$, $n(1-\mu_0) = 48 \geq 10$.
- **Sample mean count:** $\bar{x} = 282/320 = 0.88125$.

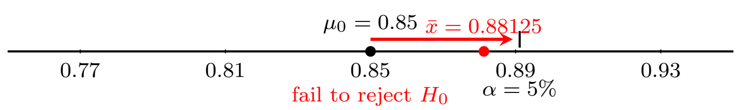
Test 1: Higher-tail test

- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.85$, $H_a\colon \mu > 0.85$.
- **Test statistic:**

$$z = \frac{0.88125 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 1.57.$$

- **p-value:** $P(Z \geq 1.57) \approx 0.0582$.
- **Decision:** Fail to reject H_0 since $0.0582 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion mean count is higher than 0.85.



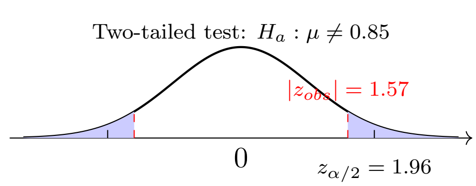


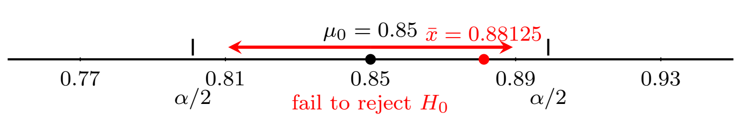
Test 2: Two-tailed test

- **Hypotheses at $\alpha = 5\%$:** $H_0\colon \mu = 0.85$, $H_a\colon \mu \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.88125 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 1.57.$$

- **p-value:** $2P(Z \geq 1.57) \approx 0.1164$.
- **Decision:** Fail to reject H_0 since $0.1164 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion mean count is different from 0.85.





12 Compliance with safety checklist (Comparison of higher-tail and two-tailed III)

A quality team states that 0.85 of shifts complete a safety checklist. In a random sample of 320 shifts, 284 completed the checklist. At $\alpha = 5\%$, test both whether the true completion mean count is higher than 0.85 and whether the true completion mean count is different from 0.85.

C5

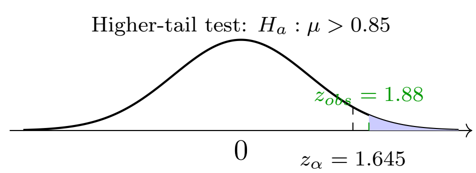
- **Parameter:** μ = true mean number of completed safety checklists per request (0 or 1).
- **Conditions:** $n\mu_0 = 272 \geq 10$, $n(1-\mu_0) = 48 \geq 10$.
- **Sample mean count:** $\bar{x} = 284/320 = 0.8875$.

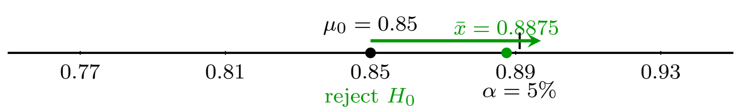
Test 1: Higher-tail test

- **Hypotheses:** $H_0\colon \mu = 0.85$, $H_a\colon \mu > 0.85$.
- **Test statistic:**

$$z = \frac{0.8875 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 1.88.$$

- **p-value:** $P(Z \geq 1.88) \approx 0.0301$.
- **Decision:** Reject H_0 since $0.0301 < 0.05$.
- **Conclusion:** There is sufficient evidence that the true checklist-completion mean count is higher than 0.85.



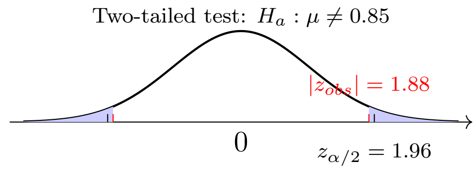


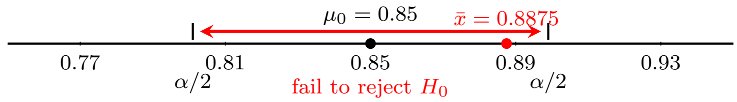
Test 2: Two-tailed test

- **Hypotheses:** $H_0\colon \mu = 0.85$, $H_a\colon \mu \neq 0.85$.
- **Test statistic:**

$$z = \frac{0.8875 - 0.85}{\sqrt{\frac{\sigma^2}{320}}} \approx 1.88.$$

- **p-value:** $2P(Z \geq 1.88) \approx 0.0602$.
- **Decision:** Fail to reject H_0 since $0.0602 > 0.05$.
- **Conclusion:** There is not enough evidence that the true checklist-completion mean count is different from 0.85.





13 On-time assignment completion mean count (higher-tail test with fixed sample mean count and variable sample size)

A school reports that students submit an average of 0.72 on-time assignments per assignment opportunity (counting variable: 1 for on time, 0 for not on time). Suppose that for samples of different sizes, the observed sample mean count is always

$$\bar{x} = 0.76.$$

At significance levels $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$, test whether the true on-time mean count is higher than 0.72, and describe how the value of the test statistic changes as the sample size n increases.

C5

- **Parameter:** μ = true mean number of on-time submissions per assignment opportunity (0 or 1).
- **Hypotheses at $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$:** $H_0\colon \mu = 0.72$, $H_a\colon \mu > 0.72$.
- **Condition:** For a one-sample mean count test, check

$$n\mu_0 = n(0.72) \geq 10 \qquad \text{and} \qquad n(1-\mu_0) = n(0.28) \geq 10.$$

  Since

$$n \geq \frac{10}{0.28} \approx 35.72,$$

  the conditions are satisfied for $n \geq 36$.
- **Sample mean count:** $\bar{x} = 0.76$.
- **Test statistic as a function of n:**

$$z(n) = \frac{0.76 - 0.72}{\sqrt{\frac{\sigma^2}{n}}}.$$

  Simplifying,

$$z(n) = \frac{0.04}{\sqrt{\frac{0.2016}{n}}} = \frac{0.04\sqrt{n}}{\sqrt{0.2016}} \approx 0.0891\sqrt{n}.$$

- **p-value as a function of n:**

$$p\text{-value}(n) = P(Z \geq z(n)).$$

  Since this is a higher-tail test,

$$p\text{-value}(n) = 1 - \Phi(z(n)),$$

  where Φ is the standard normal cumulative distribution function.

- **Interpretation of the formulas:** Since $z(n) \approx 0.0891\sqrt{n}$, the test statistic increases as n increases when the sample mean count stays fixed at 0.76. Therefore, the p-value decreases as n increases.

- **Critical values:** For a higher-tail test,

$$z_{10\%} = 1.282, \qquad z_{5\%} = 1.645, \qquad z_{1\%} = 2.326.$$

- **When does rejection begin?** Reject H_0 when $z(n) > z_\alpha$.

  At $\alpha = 10\%$,

$$0.0891\sqrt{n} > 1.282$$

  so

$$\sqrt{n} > \frac{1.282}{0.0891} \approx 14.39, \qquad n > 207.0.$$

  So rejection begins for

$$n \geq 208.$$

  At $\alpha = 5\%$,

$$0.0891\sqrt{n} > 1.645$$

  so

$$\sqrt{n} > \frac{1.645}{0.0891} \approx 18.46, \qquad n > 340.9.$$

  So rejection begins for

$$n \geq 341.$$

  At $\alpha = 1\%$,

$$0.0891\sqrt{n} > 2.326$$

  so

$$\sqrt{n} > \frac{2.326}{0.0891} \approx 26.11, \qquad n > 681.5.$$

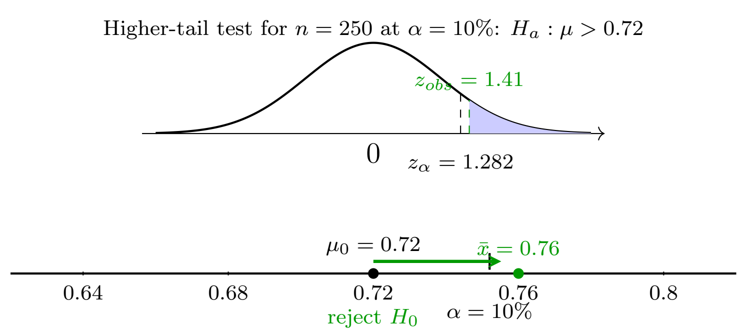
  So rejection begins for

$$n \geq 682.$$

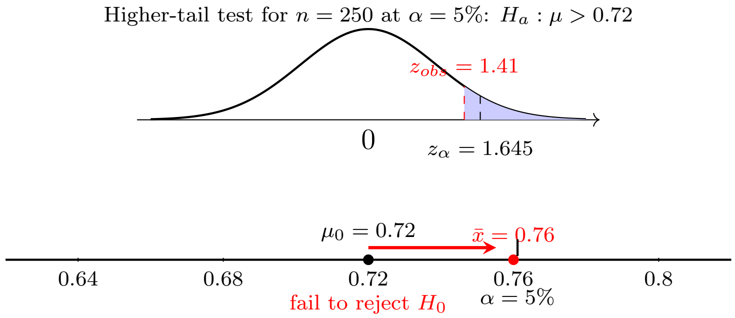
- **Sample 1:** $n = 250$.

$$z(250) \approx 0.0891\sqrt{250} \approx 1.41, \qquad p\text{-value}(250) \approx 0.0795.$$

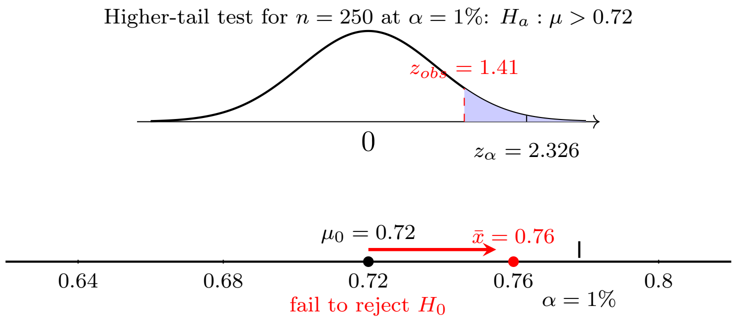
  At $\alpha = 10\%$, reject H_0 since $0.0795 < 0.10$.



  At $\alpha = 5\%$, fail to reject H_0 since $0.0795 > 0.05$.



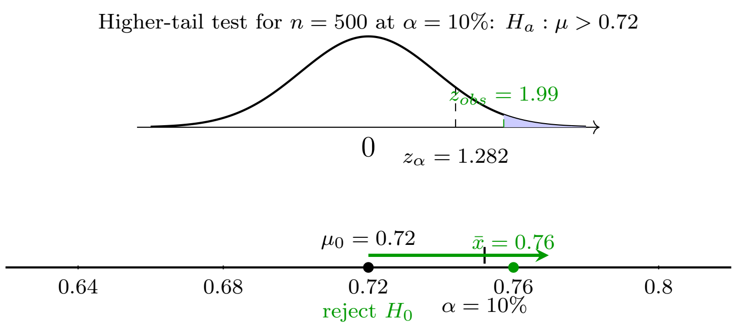
At $\alpha = 1\%$, fail to reject H_0 since $0.0795 > 0.01$.



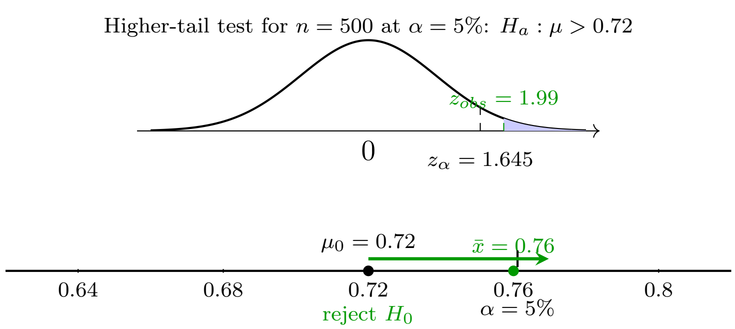
- **Sample 2:** $n = 500$.

$$z(500) \approx 0.0891\sqrt{500} \approx 1.99, \qquad p\text{-value}(500) \approx 0.0232.$$

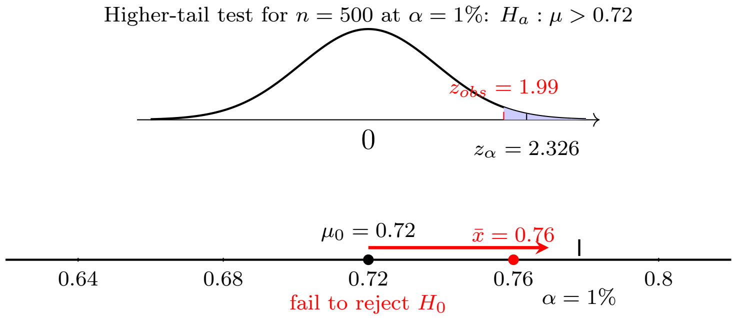
At $\alpha = 10\%$, reject H_0 since $0.0232 < 0.10$.



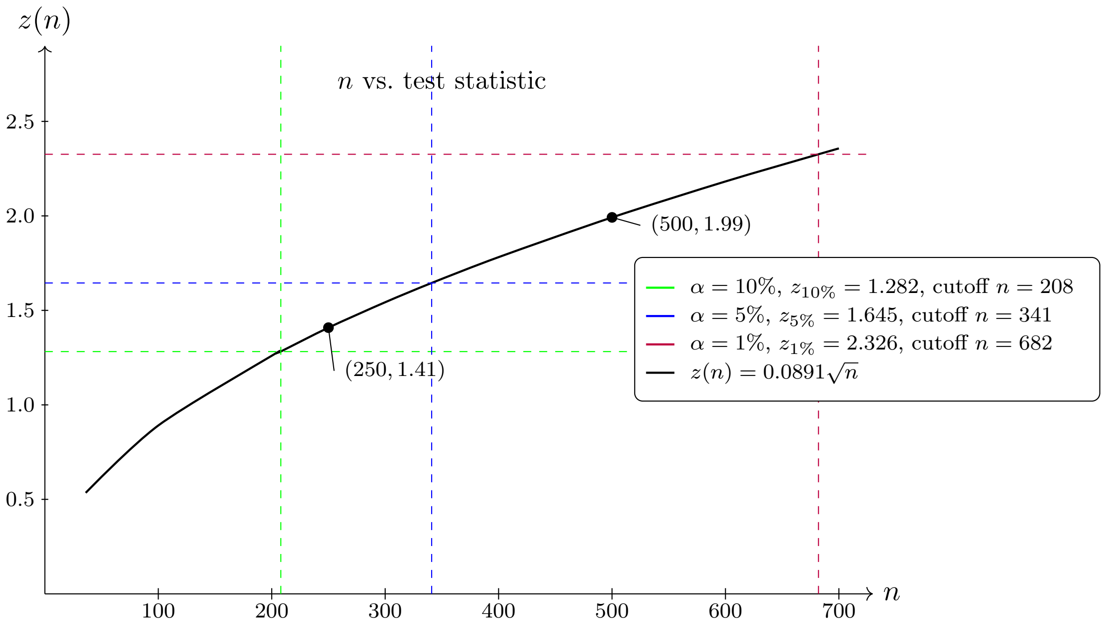
At $\alpha = 5\%$, reject H_0 since $0.0232 < 0.05$.

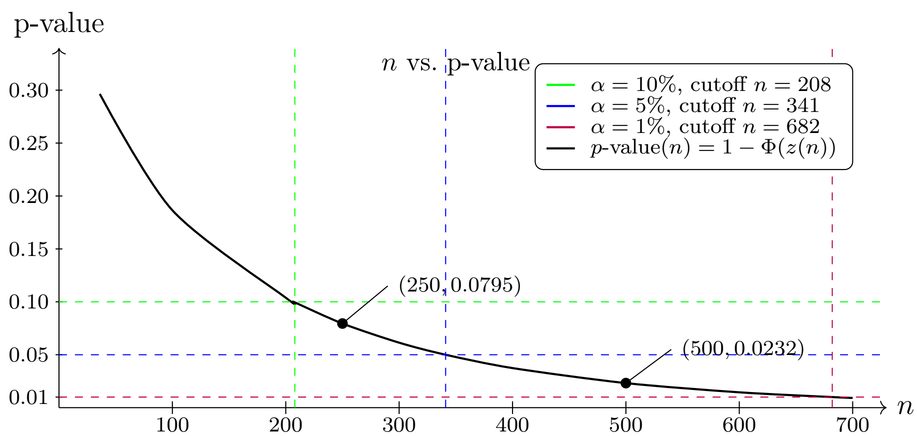


At $\alpha = 1\%$, fail to reject H_0 since $0.0232 > 0.01$.



- **Conclusion:** If the sample mean count remains fixed at 0.76, then increasing the sample size increases the test statistic and decreases the p-value. As a result, rejection of H_0 first occurs at the 10% level, then at the 5% level, and finally at the 1% level as n becomes large enough.





14 Workshop attendance rate (higher-tail test with fixed sample size and variable sample mean count)

A school reports that students attend an average of 0.72 workshops per weekly opportunity (counting variable: 1 if attended, 0 if not). Suppose that the sample size is fixed at

$$n = 400,$$

and that the observed sample mean count $\bar{x}$ varies from one sample to another. At significance levels $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$, test whether the true attendance mean count is higher than 0.72, and describe how the value of the test statistic and the p-value change as the observed sample mean count changes.

C5

- **Parameter:** μ = true mean number of workshop attendances per weekly opportunity (0 or 1).
- **Hypotheses at $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$:**

$$H_0\colon \mu = 0.72, \qquad H_a\colon \mu > 0.72.$$

- **Conditions:** For a one-sample mean count test, check

$$n\mu_0 = 400(0.72) = 288 \geq 10 \qquad \text{and} \qquad n(1-\mu_0) = 400(0.28) = 112 \geq 10.$$

  So the normal conditions are satisfied.

- **Test statistic as a function of $\bar{x}$:**

$$z(\bar{x}) = \frac{\bar{x} - 0.72}{\sqrt{\frac{\sigma^2}{400}}}.$$

  Since

$$\sqrt{\frac{\sigma^2}{400}} = \sqrt{0.000504} \approx 0.02245,$$

  we obtain

$$z(\bar{x}) \approx \frac{\bar{x} - 0.72}{0.02245} \approx 44.54(\bar{x} - 0.72).$$

- **p-value as a function of $\bar{x}$:** Since this is a higher-tail test,

$$p\text{-value}(\bar{x}) = P(Z \geq z(\bar{x})) = 1 - \Phi(z(\bar{x})).$$

- **Interpretation of the formulas:** As $\bar{x}$ increases, the quantity $\bar{x} - 0.72$ increases, so the test statistic $z(\bar{x})$ increases. Since the p-value is

$$1 - \Phi(z(\bar{x})),$$

  the p-value decreases as $\bar{x}$ increases.

- **Critical values:**

$$z_{10\%} = 1.282, \qquad z_{5\%} = 1.645, \qquad z_{1\%} = 2.326.$$

- **Rejection thresholds in terms of $\bar{x}$:** Reject H_0 when

$$z(\bar{x}) > z_\alpha.$$

At $\alpha = 10\%$,

$$44.54(\bar{x} - 0.72) > 1.282,$$

so

$$\bar{x} - 0.72 > \frac{1.282}{44.54} \approx 0.0288,$$

and therefore

$$\bar{x} > 0.7488.$$

At $\alpha = 5\%$,

$$44.54(\bar{x} - 0.72) > 1.645,$$

so

$$\bar{x} - 0.72 > \frac{1.645}{44.54} \approx 0.0369,$$

and therefore

$$\bar{x} > 0.7569.$$

At $\alpha = 1\%$,

$$44.54(\bar{x} - 0.72) > 2.326,$$

so

$$\bar{x} - 0.72 > \frac{2.326}{44.54} \approx 0.0522,$$

and therefore

$$\bar{x} > 0.7722.$$

- **Example 1:** Suppose the sample gives

$$\bar{x} = 0.7525.$$

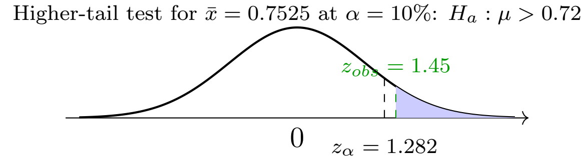
Then

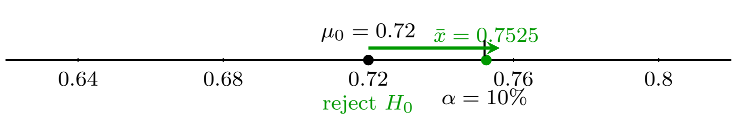
$$z(0.7525) = \frac{0.7525 - 0.72}{\sqrt{\frac{\sigma^2}{400}}} \approx 1.45.$$

The corresponding p-value is

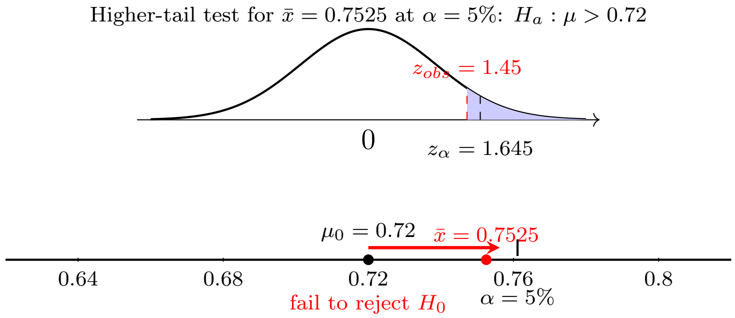
$$p\text{-value}(0.7525) = P(Z \geq 1.45) \approx 0.0739.$$

At $\alpha = 10\%$, reject H_0 since $0.0739 < 0.10$.

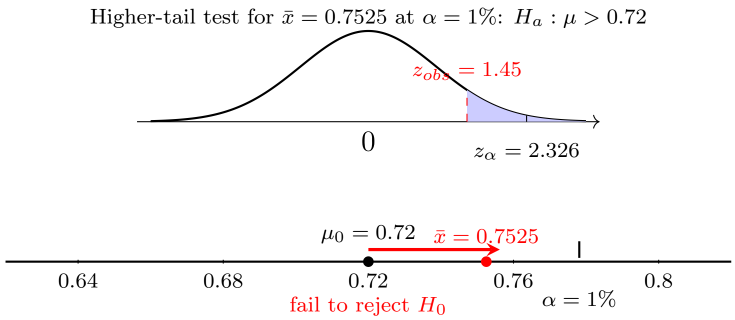




At $\alpha = 5\%$, fail to reject H_0 since $0.0739 > 0.05$.



At $\alpha = 1\%$, fail to reject H_0 since $0.0739 > 0.01$.



- **Example 2:** Suppose the sample gives

$$\bar{x} = 0.7650.$$

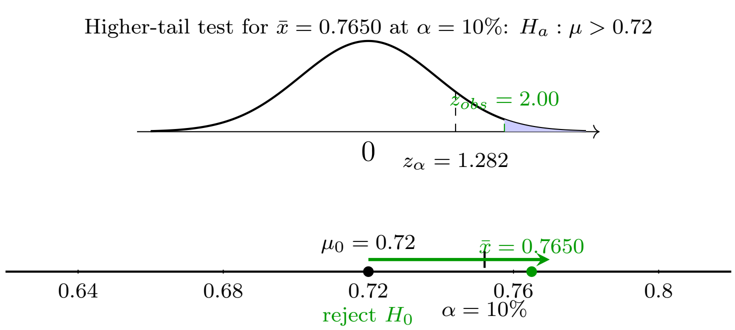
Then

$$z(0.7650) = \frac{0.7650 - 0.72}{\sqrt{\frac{\sigma^2}{400}}} \approx 2.00.$$

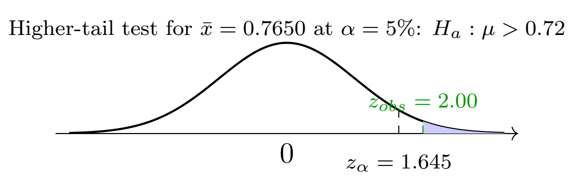
The corresponding p-value is

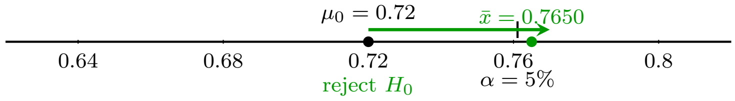
$$p\text{-value}(0.7650) = P(Z \geq 2.00) \approx 0.0225.$$

At $\alpha = 10\%$, reject H_0 since $0.0225 < 0.10$.

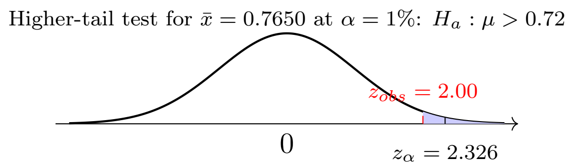


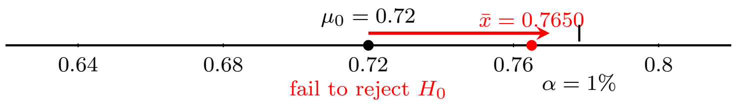
At $\alpha = 5\%$, reject H_0 since $0.0225 < 0.05$.



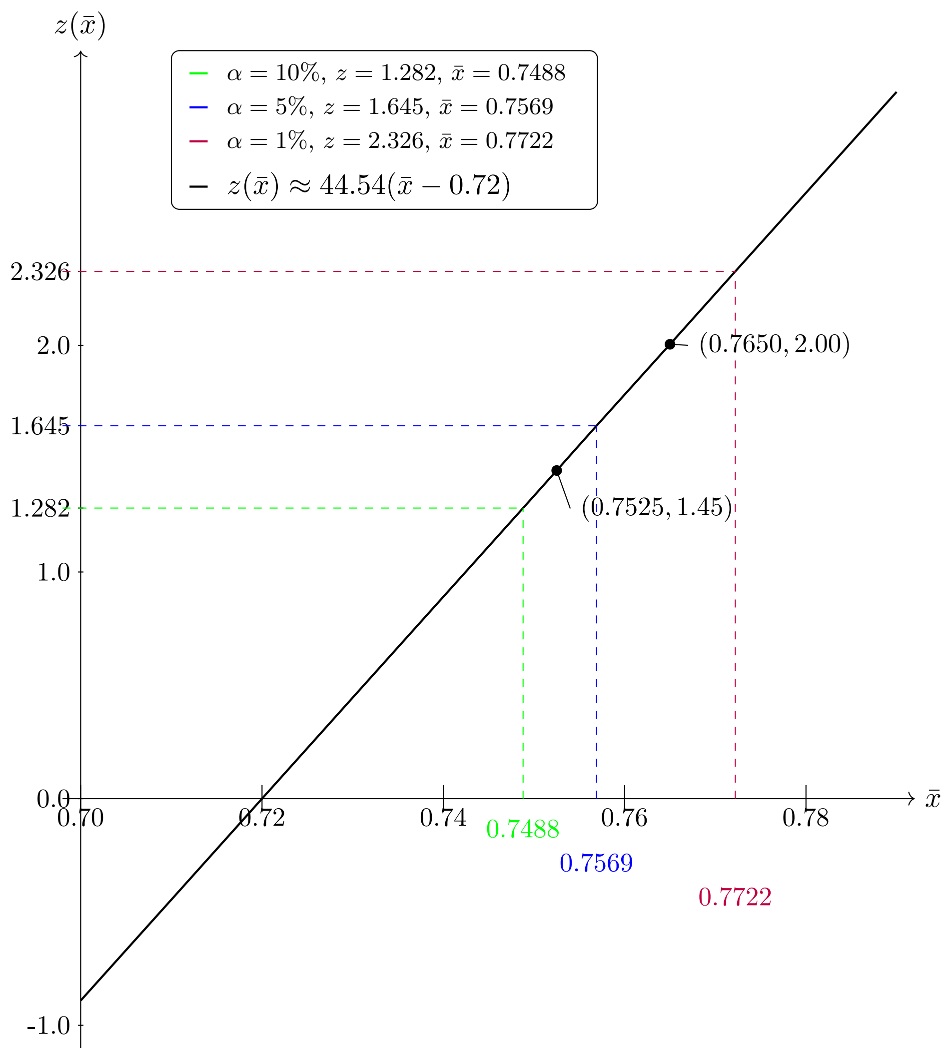


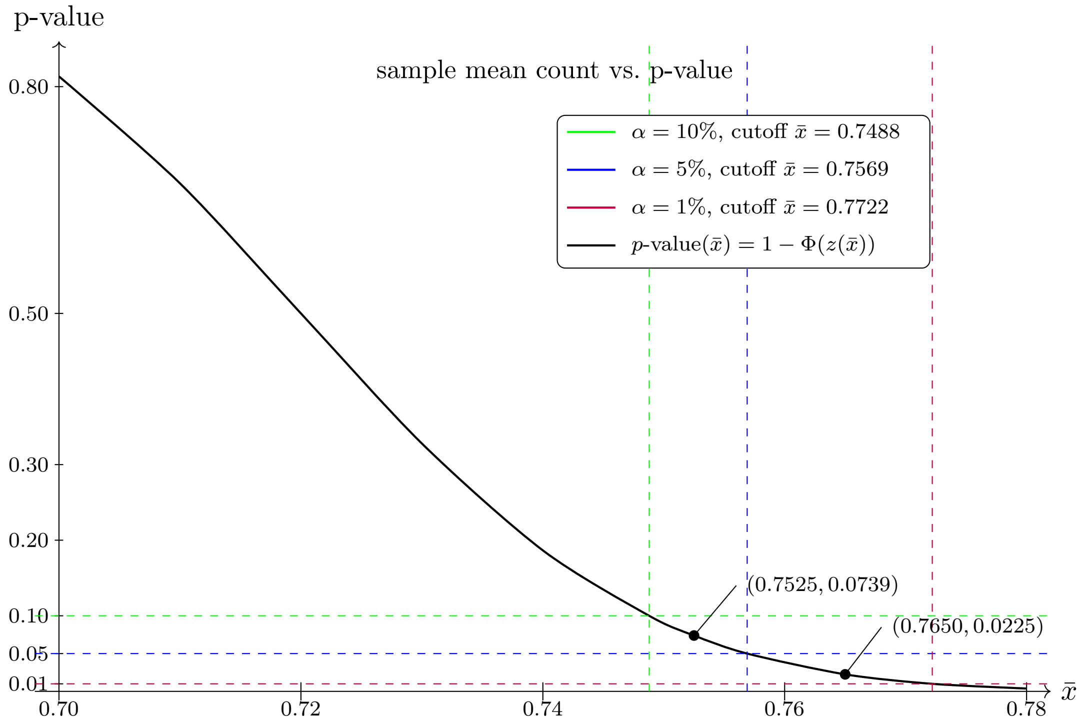
At $\alpha = 1\%$, fail to reject H_0 since $0.0225 > 0.01$.





- **Conclusion:** With the sample size fixed at $n = 400$, increasing the observed sample mean count $\bar{x}$ increases the test statistic and decreases the p-value. Therefore, rejection begins once $\bar{x}$ becomes large enough. It starts at the 10% level, then at the 5% level, and finally at the 1% level as $\bar{x}$ increases.





p-value
sample mean count vs. p-value
$\alpha = 10\%$, cutoff $\bar{x} = 0.7488$
$\alpha = 5\%$, cutoff $\bar{x} = 0.7569$
$\alpha = 1\%$, cutoff $\bar{x} = 0.7722$
p-value$(\bar{x}) = 1 - \Phi(z(\bar{x}))$
0.80
0.50
0.30
0.20
0.10
0.05
0.01
$(0.7525, 0.0739)$
$(0.7650, 0.0225)$
0.70
0.72
0.74
0.76
0.78
$\bar{x}$